

UDARA SHANUKA

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SUMMARY

Electronics and Computer Science undergraduate specializing in backend engineering, cloud infrastructure, and DevOps automation. Creator and maintainer of AxCom, an open-source headless commerce engine in Go, and builder of production systems end to end including a live commercial web application and an AI-powered booking platform using Go, Express.js, TypeScript, Kubernetes, Terraform, and GitOps CI/CD pipelines. Strong foundation in scalable system design, API-first development, and containerised deployment across AWS, GCP, and Azure.

EDUCATION

BSc (Hons) in Electronics and Computer Science

2022 - Present

University of Kelaniya, Sri Lanka

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Database Management Systems, Data Communications & Networks, Web & Internet Technologies, Cyber Security & Forensics, Full-Stack Development, Machine Learning, Intelligent Systems.

TECHNICAL SKILLS

- **Languages:** Go, JavaScript, TypeScript, Java, Python, C
- **Backend:** Node.js, Express.js, Gin, Spring Boot, REST APIs, Prisma ORM, Redis, RabbitMQ
- **DevOps & Cloud:** Docker, Kubernetes, Helm, ArgoCD, GitHub Actions, Jenkins, Terraform, Ansible, OpenTelemetry, Prometheus, Grafana, AWS, GCP, Azure, Linux
- **Frontend:** Next.js, React, Angular, Zustand, TanStack Query
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Tools:** Git, Postman, Jest, ESLint, Husky, IntelliJ, VS Code

OPEN SOURCE

AxCom - Modular Headless Commerce Engine (Creator & Maintainer) | [GitHub](#)

2026 – Present

Go, Gin, PostgreSQL, MongoDB, Redis, RabbitMQ, OpenTelemetry, Docker, GitHub Actions, Docusaurus

Open-source (Apache-2.0) headless e-commerce backend in Go - axcom.axiolon.com | Docs: axiolon.github.io/axcom

- Designed and built a headless commerce engine with seven independent domain modules (auth, catalog, inventory, cart, orders, payments, shipping) following clean architecture with zero cross-module imports.
- Implemented swappable infrastructure adapters, PostgreSQL/MongoDB storage, Redis caching, RabbitMQ event messaging, and S3/R2 file storage configurable without changing business logic.
- Built JWT authentication with role-based access control, rate limiting, and CORS middleware at the API gateway layer.
- Instrumented the engine with OpenTelemetry tracing, Prometheus metrics, and structured JSON logging for Grafana/Jaeger-compatible observability.
- Set up GitHub Actions CI/CD with automated releases and Docker image publishing; ships as a single static binary with a Docker Compose local environment.
- Authored a full documentation site (Docusaurus) with API reference, module guides, and contribution, security, and code-of-conduct policies for community collaboration.

PROJECTS

TrekDesk AI - AI-Powered Tour Booking Assistant | [Live Site](#)

Mar 2026

Node.js, Express, TypeScript, React, WebSockets, PostgreSQL, pgvector, Gemini Live API, Google APIs

- Built the backend for an AI-driven tour booking assistant supporting real-time chat and voice interactions.
- Developed a RAG pipeline with PostgreSQL and pgvector for semantic search over tour and booking data.
- Integrated Gemini Live API over WebSockets for low-latency conversational AI (text and voice).
- Implemented AI tool-calling for dynamic actions such as availability checks and booking workflows.
- Integrated Google Calendar API for automated scheduling and guide availability management.
- Built secure Google OAuth authentication with JWT-based sessions, plus an embeddable widget with domain validation for third-party sites.

University Multi-Portal Management System - SSO + DevOps Deployment | [GitHub](#) Dec 2025 - Jan 2026

Next.js, Turborepo, TypeScript, Prisma, PostgreSQL, K3s, Terraform, ArgoCD, Ansible, WSO2 Identity Server

- Built a multi-portal platform (Student, Library, Admin) with a shared component architecture using Next.js and Turborepo.
- Implemented SSO and role-based access control with WSO2 Identity Server.
- Designed scalable backend services with TypeScript, Prisma, and PostgreSQL for multi-tenant data management.
- Automated Azure infrastructure provisioning with Terraform and deployed workloads on a K3s cluster.
- Established a GitOps CI/CD pipeline with ArgoCD and Ansible for zero-downtime deployments, monitored with Prometheus and Grafana.

Cloud-Native API Platform with WSO2 API Management | [GitHub](#)

Nov 2025

Node.js, Kubernetes, Docker, Helm, WSO2 API Manager, OpenAPI 3.0, MySQL

- Built cloud-native backend APIs for e-commerce domain services (catalog, cart, orders) with an API-first OpenAPI 3.0 design.
- Integrated WSO2 API Gateway for OAuth2/JWT authentication, rate limiting, and policy enforcement.
- Deployed containerised services on Kubernetes with Helm, using persistent volumes for database storage.

Trekking Tours - Commercial Web Application | [Live Site](#)

Feb 2026

Next.js, Vercel, Resend, Analytics

- Developed and deployed a production website serving real customer inquiries.
- Delivered an SEO-optimised, responsive UI and integrated analytics to track visitor behaviour and improve engagement.

AI-Driven 3D Posture Detection in Olympic Weightlifting (Research) | [GitHub](#)

Jun 2024 – Jun 2025

Python, TensorFlow, OpenCV, MediaPipe, LSTM

- Designed an automated multi-camera video pipeline covering frame extraction, pose estimation, and joint-angle computation.
- Built Python feature-engineering pipelines to automate dataset creation from raw video.
- Developed CNN-LSTM models for phase classification and real-time feedback in Snatch and Clean-and-Jerk lifts.

RosaLover - E-Commerce Platform (Group Project) | [GitHub](#)

Oct 2024 – Jan 2025

React, Spring Boot, MySQL, Stripe

- Developed an e-commerce site with product management, wishlist, cart, and Stripe payment integration.
- Built an admin panel for inventory, orders, and user-role management; contributed across frontend, backend, and integration.

CERTIFICATIONS

- Front End Development Libraries - freeCodeCamp (Aug 2024)
- Server-Side Web Programming - University of Moratuwa (Jul 2024)
- Front End Web Development - University of Moratuwa (May 2024)
- Web Design for Beginners - University of Moratuwa (Jan 2024)
- JavaScript Algorithms and Data Structures - freeCodeCamp (Dec 2023)
- Responsive Web Design - freeCodeCamp (Aug 2023)

References available upon request.